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# The paleomagnetism and cooling history of metamorphic and igneous rocks from the Limpopo Mobile Belt, southern Africa

GERALD E. MORGAN* *Department of Earth Sciences, University of Leeds, Leeds LS2 9JT, United Kingdom*

## ABSTRACT

**The Limpopo Mobile Belt is a zone of predominantly high-grade gneisses located between the Rhodesian and Kaapvaal cratons in southern Africa. The last major period of metamorphism and deformation in the Belt occurred ~2600 m.y. ago, but numerous Rb-Sr mineral ages indicate that final cooling occurred at 2000 to 1950 m.y. B.P. Paleomagnetic investigations have been carried out on samples from 78 sites in the metamorphic rocks and 6 sites in post-tectonic dikes that intruded at 1876 (±68) m.y. B.P. Detailed demagnetization of the metamorphic sites revealed three stable magnetizations. Component "A" (D = 352°, I = 60°, $\alpha_{95}$ = 7°; corresponding pole 26°N, 22°E) is thought to have been acquired during cooling of the Belt at ~1980 m.y. B.P., and, together with a result previously published, suggests diachronous cooling across the Belt with a polarity change occurring during the cooling interval. Component "B" (D = 41°, I = -7°, $\alpha_{95}$ = 16°; corresponding pole 46°N, 99°E) is probably a later chemical remanent magnetization (CRM), but its exact age is unknown. Component "C" (D = 357°, I = -66°, $\alpha_{95}$ = 10°) is very close to the Earth's present field direction and is probably of recent origin. The post-tectonic dikes yielded a stable magnetization: D = 318°, I = 73°, $\alpha_{95}$ = 6° and corresponding pole 3°N, 9°E and is thought to have been acquired at the time of intrusion at ~1876 m.y. B.P. The new data assist in refining the apparent polar wander (a.p.w.) path for the Kalahari Shield for the interval circa 2050 to circa 1800 m.y. ago. The data also suggests that over this time interval (a) it is difficult to draw firm conclusions about the relative positions of the various African shields and the South American Guayana Shield; and (b) the paleomagnetic data argue against any proposed supercontinent reconstruction of Africa and North America, and that at least the Kalahari Shield was in strong relative motion with respect to North America.**

## INTRODUCTION

This paper reports paleomagnetic investigations in the Precambrian Limpopo Mobile Belt in southern Africa. The major part of the work consisted of a study of high-grade metamorphic rocks collected from 78 sites across the Belt, the object being that of identifying a regional cooling magnetization corresponding to the widespread (circa 1980-m.y.) mineral ages found in the Belt. A smaller study involved 6 sites in a swarm of post-tectonic dikes that intruded into the Belt ~100 m.y. after the regional cooling event. A brief account of some preliminary results from the metamorphic rocks of the Belt has been published elsewhere (Morgan and Briden, 1981). Apart from a study by Jones and others (1975) on seven sites in the extensions of the Great Dyke and its satellites from the northern part of the Belt in Zimbabwe (Rhodesia), this is the first paleomagnetic study carried out in the Limpopo Belt.

## GEOLOGY

The Limpopo Belt (Fig. 1) is a linear zone ~600 km long and 240–320 km wide of intensely deformed, high-grade metamorphic rocks situated between the Archean Rhodesian and Kaapvaal cratons (Mason, 1973; Ermanovics and others, 1977). The predominant lithologies present are tonalitic and granitic gneisses, but there are also subordinate amounts of metasediments and basic and ultrabasic intrusives and extrusives which have been infolded and, together with the basement, subjected to amphibolite- or granulite-facies metamorphism. The Limpopo Belt has been divided into a Central Zone where both amphibolite- and granulite-facies rocks are present, separated by major shear belts from Northern and Southern Marginal Zones which have predominantly granulite-facies assemblages. In both the north and the south, there is a gradual transition from the high-grade metamorphic rocks of the marginal zones to the granite-greenstone rocks of the cratons. Similarly, there is a gradual transition in structural features, and the east-northeast trend characteristic of the Belt is still apparent in the southern part of the Rhodesian craton and in the northern part of the Kaapvaal craton. To the east and west, the Belt disappears under a cover of younger rocks, mainly those of the Karroo Supergroup (Permian-Jurassic).

Apart from the post-Karroo ring complexes of the Nuanetsi igneous province at the eastern end of the Belt, the only post-tectonic intrusions present are minor swarms of dolerite dikes. There is evidence of persistent later activity along old fault lines with predominantly vertical movements. Within the belt are two predominantly fault-bounded basins; the Tuli Trough contains Karroo sediments and volcanics, and the Soutpansberg Trough contains sediments and volcanics of both the Soutpansberg Group, the deposition of which commenced at ~1770 m.y. B.P. (Barton, 1979), and the Karroo Supergroup.

## GEOCHRONOLOGY

The Limpopo Belt has been the subject of numerous geochronological investigations, and ~25 Rb-Sr whole-rock isochrons and errorchrons have been obtained from metamorphic rocks in various parts of the Belt (Van Breemen

*Present address: Exploration Department, Gulf Canada Resources Inc., 401 9th Avenue S.W., Calgary, Alberta, Canada, T2P 2H7.

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and Dodson, 1972; Hickman and Wakefield, 1975; Barton and Ryan, 1977; Barton, 1979). The majority of these ages fall in the range circa 2750 to circa 2450 m.y. and are thought to date the last major period of widespread metamorphism and deformation in the Belt. (All Rb-Sr ages quoted in this paper have been calculated, or, if necessary, recalculated using the new decay constant for $^{87}Rb$ of $1.42 \times 10^{-11}y^{-1}$.) A few older ages, ranging as much as circa 3775 m.y., have been obtained from two areas close to Messina. These apparently were not affected by the later event, and three younger ages of 2193, 1904, and 1790 m.y. have also been obtained from metamorphic rocks of the Belt. The significance of these younger ages is discussed below. More important, from the paleomagnetic point of view, however, is the fact that 30 Rb-Sr mineral, mineral-pair, mineral–whole-rock, and thin-slice isochron ages from metamorphic rocks in the Belt show a very pronounced peak at 1950–2000 m.y. (Fig. 2). These mineral ages have been obtained from widely separated localities across the Belt but not from the Southern Marginal Zone, where mineral dating has yet to be carried out. Whether these circa 1980-m.y. mineral ages represent final uplift and cooling after a prolonged period of deep burial, or whether they represent a later thermal event distinct from an earlier cooling episode is not known. What is important is that the region last cooled through the blocking temperatures for Rb-Sr mineral systems at ~1950–2000 m.y. B.P., and the main objective of the paleomagnetic study was to identify and investigate the magnetization acquired during this cooling event. No later events other than slight tilting, block faulting, deposition and erosion of thin cover, and the intrusion of minor dike swarms are known, so that the Limpopo Belt offered an ideal situation for the paleomagnetic study of regional cooling free from obvious severe disturbing influences.

Of the four youngest mineral ages (1763, 1800, and 1800 m.y.), three were all obtained from gneisses at a single locality near Tshipise ~30 km south-southwest of Messina, the same locality that yielded the unusually low Rb-Sr whole-rock age of 1790 m.y. mentioned above (Barton, 1979). These gneisses were sampled at a point very close to the edge of the circa 1770-m.y. Soutpansberg Trough, and Barton suggested that the circa 1800-m.y. event that was recorded in the gneisses at this locality was related to the initial formation of the Trough and was restricted to the area immediately around its margins. Of the remaining mineral ages, there appears to be no clear relationship between position in the Belt and age; however, the three youngest ages were obtained from the three localities in the northern part of the Belt on its border with the Rhodesian craton in the region of the Greak Dyke extensions. In fact, the mean of the 5 mineral ages from this northern area is 1895 m.y. (S.D. = 73 m.y.), whereas the mean of all the remaining mineral ages (omitting the 3 anomalously low ages from Tshipise) is 1980 m.y. (S.D. = 39 m.y.), suggesting that the northern area cooled significantly later than did the main part of the Belt. There was also the interesting possibility that this difference in cooling ages between two areas of the Belt might be detected paleomagnetically.

A whole-rock age of 2193 m.y. from gneisses in the Mahalapye area was interpreted by Van Breemen and Dodson (1972) as representing partial homogenization during a remobilization episode at ~1960 m.y. that is, at the same time as the thermal event which produced the widespread mineral ages. A whole-rock age of 1904 m.y. from gneisses ~45 km west-northwest of Messina was also interpreted by Barton and Ryan (1977) as reflecting the same event. Post-tectonic dolerite dikes from the Messina region of the Belt and from the Southern Marginal Zone have yielded Rb-Sr whole-rock isochrons of 2216, 1905, and 1876 m.y. and are thought to date the time of emplacement of these intrusions. Six closely adjacent dikes from the swarm which yielded the youngest age were the subject of part of the paleomagnetic study reported here.

## THE SAMPLING SITES

The Limpopo Belt straddles the boundaries between South Africa, Botswana, and Zimbabwe. Approximately one-third of the Belt lies in each country. The sampling sites (Fig. 3) were unfortunately confined to South Africa and Botswana, as Zimbabwe (then Rhodesia) was inaccessible for field work throughout the period of the study. Altogether, 78 sampling sites were obtained in metamorphic rocks, and 6 sites were obtained from post-tectonic dikes near Messina (see below). The metamorphic sites were spread across the entire width of the Belt and along as much of the length of the Belt as the inaccessibility of Zimbabwe permitted.

The majority of sampling sites were located in the granitic/tonalitic gneiss, which is the predominant lithology in the Belt, but some were located in more mafic units, and a few sites undoubtedly were in metasedimentary units, such as calc-silicates. Six sites were also located in the predominantly meta-volcanic Tati Schist Relict in the area around Francistown on the border of the Rhodesian craton. The most northerly and southerly of the sampling sites were located in little-altered igneous intrusions of the Rhodesian and Kaapvaal cratons, respectively. Site 58 was located in the diorite ~6 km north of Francistown, and site 86 was located in the Matok Granite Pluton ~55 km south-southeast of Louis Trichardt. At many sites, no foliation was

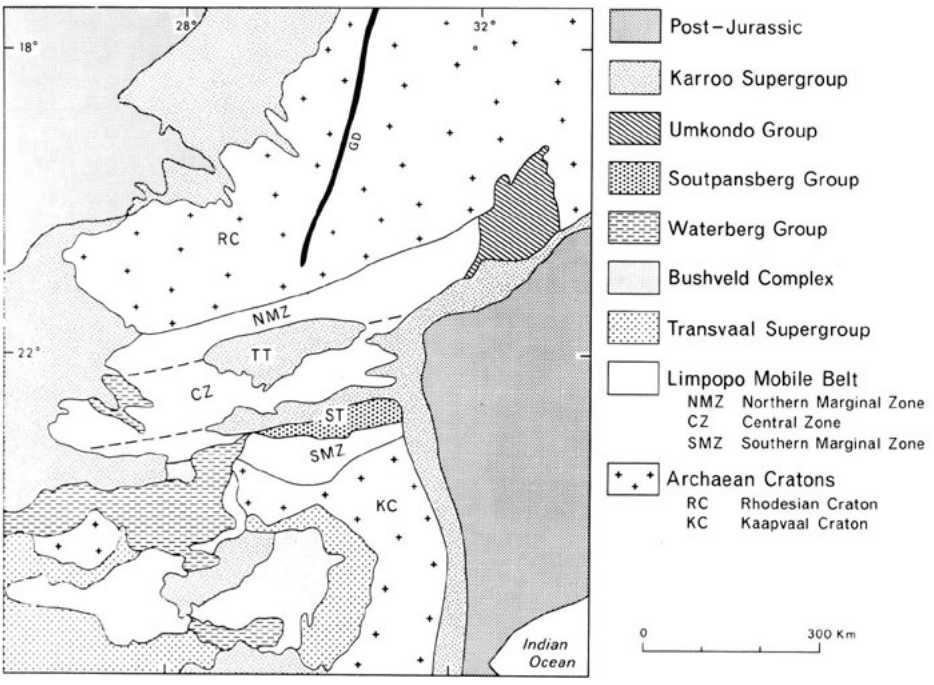
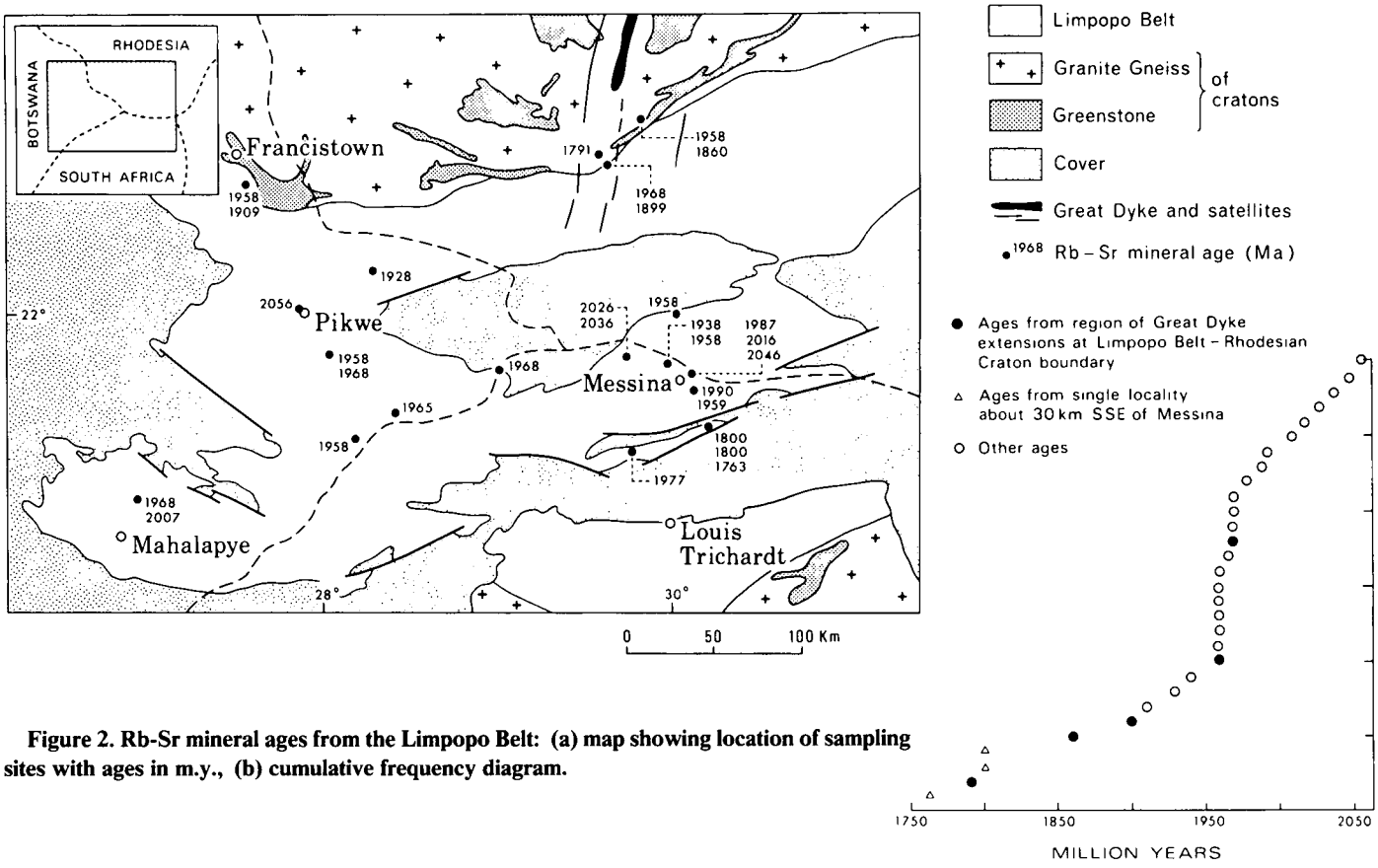



**Figure 1. Generalised geologic map showing the regional setting of the Limpopo Mobile Belt. GD, Great Dyke; TT, Tuli Trough; ST, Soutpansberg Trough.**

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a




b

**Figure 2. Rb-Sr mineral ages from the Limpopo Belt: (a) map showing location of sampling sites with ages in m.y., (b) cumulative frequency diagram.**

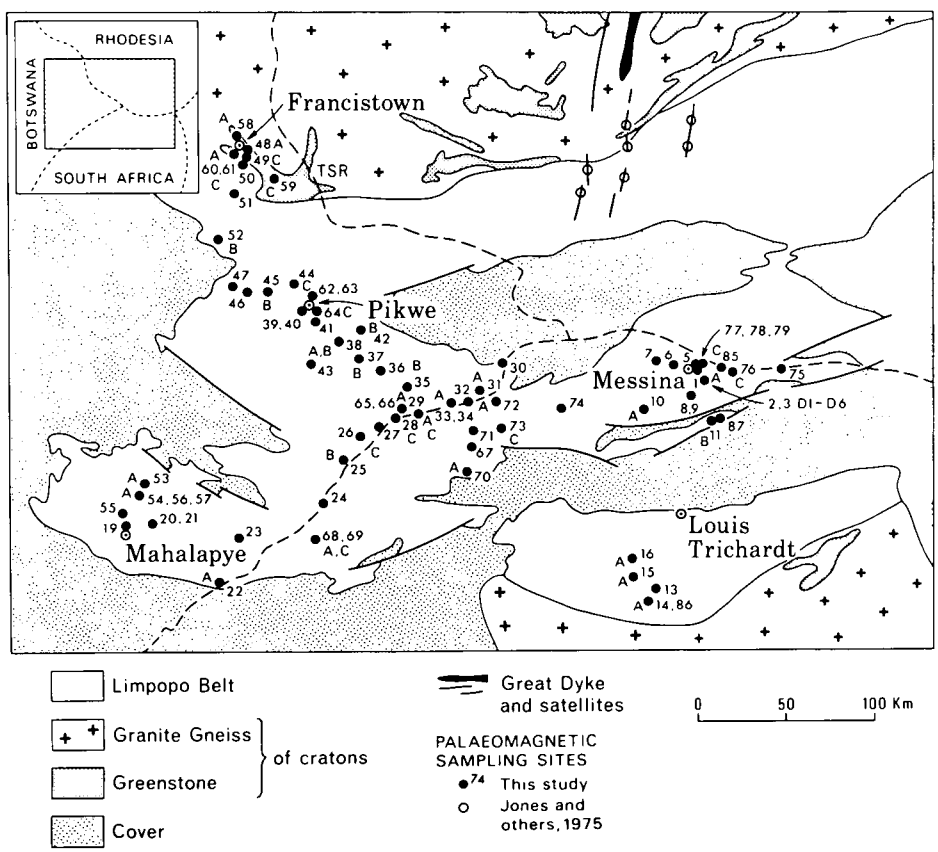



**Figure 3. Map of the Limpopo Belt showing locations of the 78 paleomagnetic sampling sites in the metamorphic rocks (numbered) and also the 6 sites in the post-tectonic dolerite dikes near Messina (labeled D1–D6). Locations of the seven sites in the extensions of the Great Dyke and its satellites into the Limpopo Belt obtained by Jones and others (1975) are also shown. Sites where "A," "B," and "C" magnetizations (see text) were found are indicated by appropriate letters. TSR, Tati Schist Relict.**

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evident, and at those sites where a slight fabric was apparent, it was generally due to compositional banding rather than to preferred mineral orientation. Apart from three underground sites (77–79) in Artonvilla Mine, Messina, where block samples were taken, all sampling was done with a portable drill. At each site, between 8 and 12 separately oriented 6–10-cm-long cores were obtained, usually spaced ~1 m apart.

## PALEOMAGNETIC RESULTS FROM THE METAMORPHIC ROCKS

### Introduction

Detailed pilot demagnetization studies indicated that alternating field (a.f.) demagnetization was much more effective at removing secondary components than was thermal demagnetization. Therefore, at the majority of sites, most of the samples were subjected to a.f. demagnetization, and only one or two samples were thermally demagnetized.

At 2 of the 78 sites, intensities of magnetization were too low for reliable measurement. At another 27 sites, the total natural remanent magnetization (NRM) directions were widely scattered, and demagnetization did not produce any appreciable improvement in grouping. At the remaining 49 sites, however, either the total NRM directions were widely scattered, but the grouping improved considerably after demagnetization, or the total NRM directions were already moderately or well grouped, and demagnetization generally, but not always, improved this grouping. These 49 sites are considered to have yielded paleomagnetically useful site mean directions of magnetization (Fig. 4). At about a third of these sites, it was found that a minority of samples behaved differently from the rest; either they were stably magnetized in a direction different from that of the majority, or they were unstably magnetized when the rest of the samples were stable. These samples were excluded from the statistical analysis of these sites. The anomalous samples did not otherwise appear to be different from the other samples, and the cause of this behavior is not understood. The number of rejected samples at sites so affected is indicated in Appendix 1.

The majority (41) of significant site mean directions fall into one of 3 fairly well defined groups of directions labeled "A," "B," and "C" in Figure 4, and 11 site mean directions fall outside these groups. (These two numbers add up to 52, not 49, because one site contains both "A" and "B" components, one site contains both "A" and "C" components, and another site contains both normal and reverse "A" components.) Paleomagnetic statistics and brief lithological details for each of these 52 sites are given in Appendix 1, and over-all statistics for the "A," "B," and "C" magnetizations are given in Table 1. The geographical distribution of the sites yielding the "A," "B," and "C" magnetizations is shown in Figure 3.

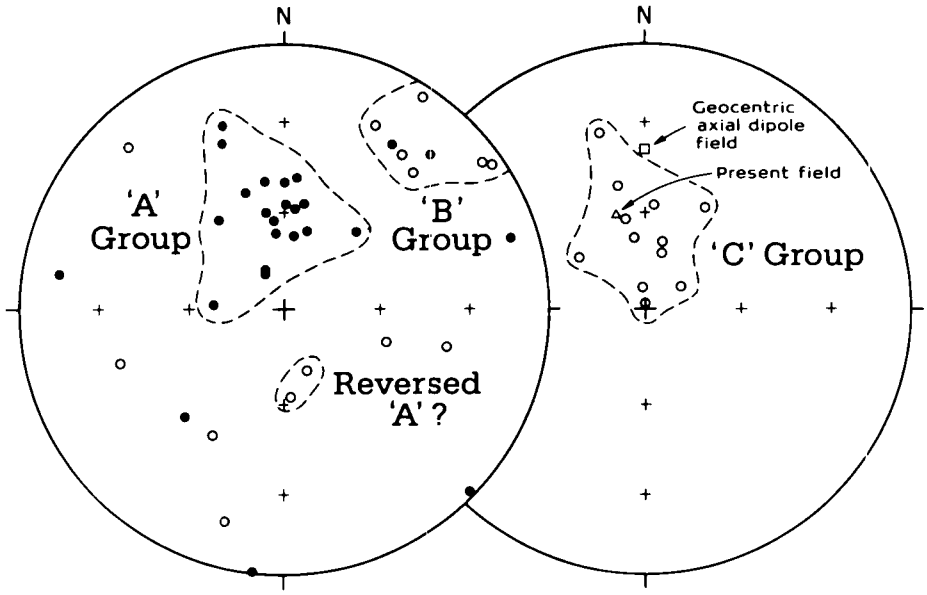



**Figure 4. Significant cleaned site mean directions from Limpopo Belt metamorphic rocks. "C" group magnetizations are shown on a separate plot for clarity. Equal area projections: solid circles = lower hemisphere, open circles = upper hemisphere.**

### The "A" Directions

The "A" direction of magnetization (northerly declination and generally moderate positive inclination) was found at 20 sites in a variety of lithologies from locations spread throughout the area of the Belt studied, including two sites (48 and 60) in the meta-volcanics of the Tati Schist Relict near Francistown. The "A" direction of magnetization was also found at two sites (58 and 86) in the little-altered igneous intrusions from the craton margins. At two sites (16a and 53), reverse "A" directions were found, although, of course, it is not possible to be completely confident that these were not part of the population of sites with random mean directions.

At many of these sites, for example, site 33 (Fig. 5), the total NRM directions were widely scattered, but a.f. demagnetization, generally at moderate to high fields, caused the vectors to move to stable end points in the "A" direction. Thermal demagnetization of samples with scattered directions, however, generally either produced no appreciable movement of the magnetization or produced only a small, partial movement toward the "A" direction (Fig. 5f). Some sites, for example, 32 and 48, had total NRM directions which were well grouped close to the "A" direction, and samples from these sites were generally very stable to both a.f. and thermal demagnetization (Fig. 5). The blocking temperatures of the grains carrying the "A" remanence were predominantly in the range 540–600 °C, but occasionally the range extended to as much as ~650 °C, and a few samples contained "A" components with blocking temperatures ranging down below 100 °C.

At the majority of sites, the individual samples reached good, stable end points during

TABLE 1. GROUP MEAN MAGNETIZATIONS FROM THE LIMPOPO BELT

| | Direction | | | | | Paleomagnetic pole |
|---|---|---|---|---|---|---|
| | N | D | I | K | $\alpha_{95}$ | |
| 1. METAMORPHIC ROCKS | | | | | | |
| Group "A"* | 21 | 352.1° | 60.0° | 23.1 | 6.7° | 26.1°N, 22.3°E (dp = 7.9°, dm = 10.3°) |
| Group "B" | 8 | 41.1° | −6.5° | 12.5 | 16.2° | 45.7°N, 99.2°E (dp = 8.1°, dm = 16.2°) |
| Group "C" | 12 | 357.2° | −66.4° | 19.4 | 10.0° | 63.5°N, 213.1°E (dp = 13.4°, dm = 16.3°) |
| 2. SAND RIVER DYKES† | 6 | 318.0° | 72.5° | 123.8 | 6.0° | 2.5°N, 92°E (dp = 9.6°, dm = 10.7°) |

*Note:* N is the number of sites; D and I are the declination and inclination of the mean direction; k is Fisher's precision parameter; $\alpha_{95}$ is the radius of 95% confidence about the mean direction; dp and dm are the minor and major semi-axes, respectively, of the oval of 95% confidence about the paleomagnetic pole.

*With sites 16(a) and 53 reversed.

†With site D6 reversed.

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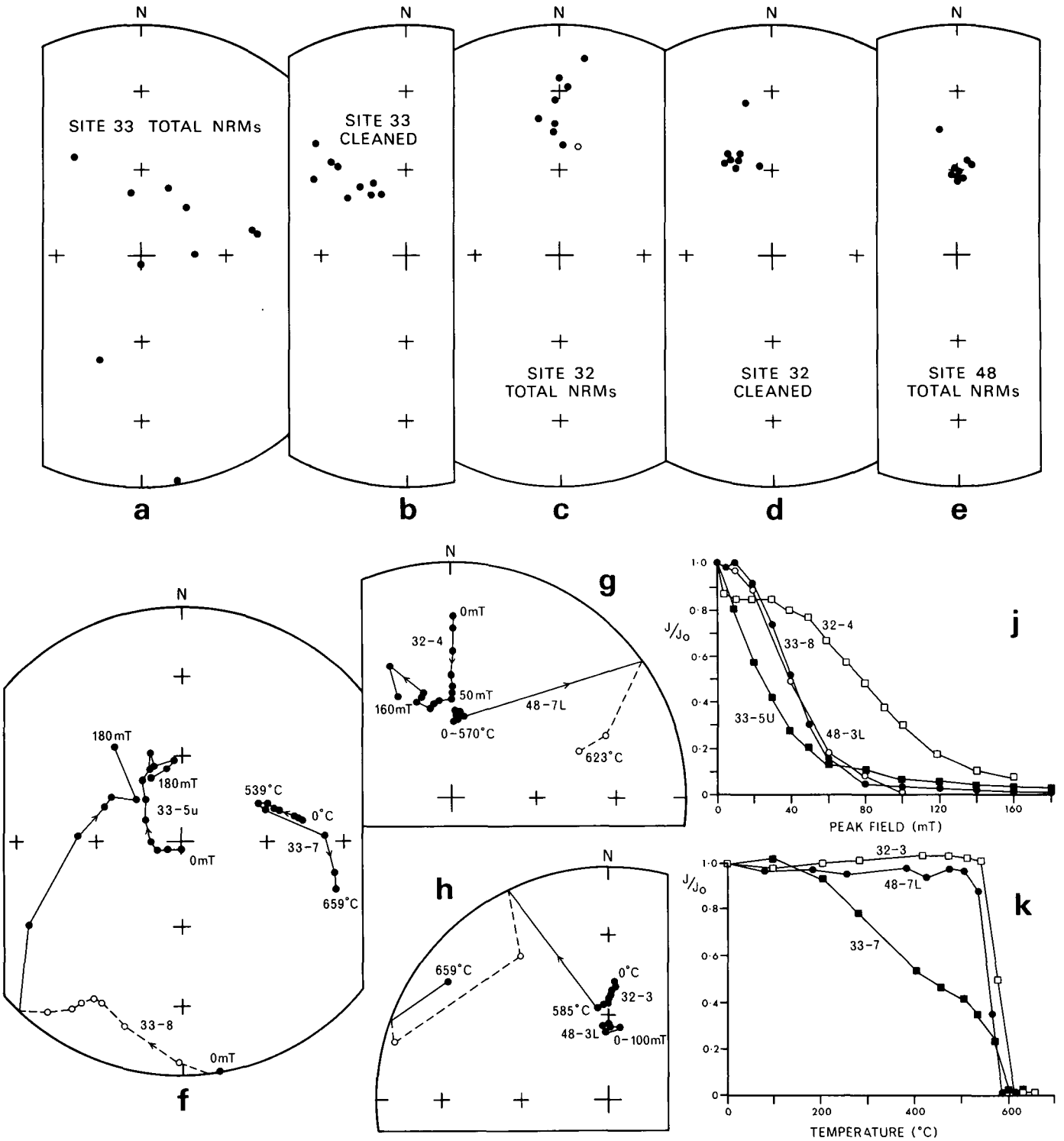



**Figure 5. Sites yielding "A" magnetizations: (a) site 33 total NRM directions; (b) site 33 after cleaning at 120–180 mT; (c) site 32 total NRM directions; (d) site 32 after cleaning at 90 mT (8 samples) and 585 °C (1 sample) with one sample rejected; (e) site 48 total NRM directions; (f) to (k) detailed a.f. and thermal demagnetization of samples from sites 33, 32, and 48. All projections are equal area: solid circles = lower hemisphere, open circles = upper hemisphere.**

a.f. demagnetization, and the site mean directions were computed by combining the end-point directions of the individual samples, including, where appropriate, the end-point direction of any sample treated thermally.

### The "B" Directions

The "B" direction of magnetization, with northeast declination and shallow positive or negative inclination, was found at eight sites, mainly from the central region of the Belt. At all of these sites, the total NRM directions were already moderately or well grouped in the "B" direction, and at only three sites did a.f. demagnetization improve the NRM grouping. At the

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majority of sites, for example, site 52 (Figs. 6a to 6e), the magnetization showed excellent directional stability to high temperatures during thermal demagnetization but only moderate stability to a.f. demagnetization, although the average directions of magnetization did not change appreciably. (Site 52 also illustrates the phenomenon, found at a number of sites, that a minority of samples at a site have directions different from that of the remainder. Six samples form a very tight group; three samples have quite different directions. On a.f. demagnetization, two of the anomalous samples showed very good directional stability in their NRM directions, but one sample moved to a typical "B" direction after 30 mT.) Most "B" sites showed good stability to thermal demagnetization, but two of them, sites 11 and 37, showed reversals during thermal demagnetization (Figs. 6f and 6h). Between the total NRM stage and 562 °C, the samples showed very good directional stability, and their intensities were reduced to ~10% of their NRM values. Between 562 and 598 °C, however (that is, on passing through the magnetite Curie point), the intensities suddenly dropped to ~1%, and there was a spectacular change in direction, corresponding closely to a 180° reversal. Over the four steps between 598 and 691 °C, the intensities remained fairly level, and the directions also stayed fairly constant in the reversed positions. Detailed a.f. demagnetization of two samples from site 37 also produced the same sudden reversal of direction, the reversal occurring at the 60/80-mT step in sample 37-9U (Figs. 6g and 6j) and at the 100/120-mT step in another sample. One sample from site 11, 11-4U, showed a double reversal during a.f. demagnetization (Figs. 6g and 6j).

The "B" component resides in grains with a wide spectrum of blocking temperatures, from <100 °C to as much as 650 to 690 °C. Generally, the bulk of the remanence resided in grains with blocking temperatures between ~100 and 580 °C. Smaller, but nevertheless significant contributions (in two cases, in the reverse direc-

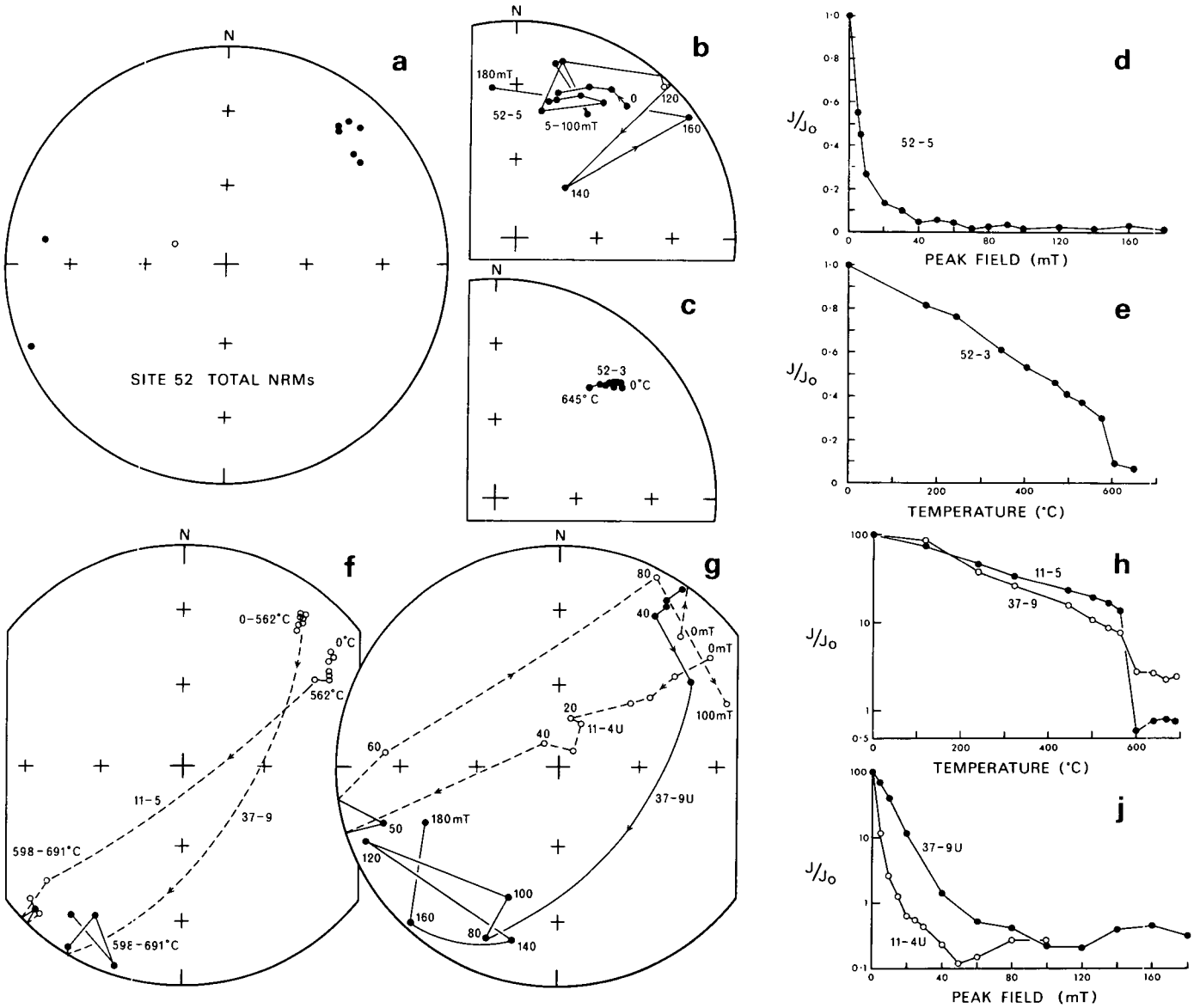



**Figure 6. Sites yielding "B" magnetizations: (a) site 52 total NRM directions, (b) to (e) detailed a.f. and thermal demagnetization of samples from site 52, (f) to (j) detailed thermal and a.f. demagnetization of samples from sites 11 and 37 showing reversals during demagnetization. Note logarithmic intensity scales in (h) and (j). All projections are equal area: solid circles = lower hemisphere, open circles = upper hemisphere.**

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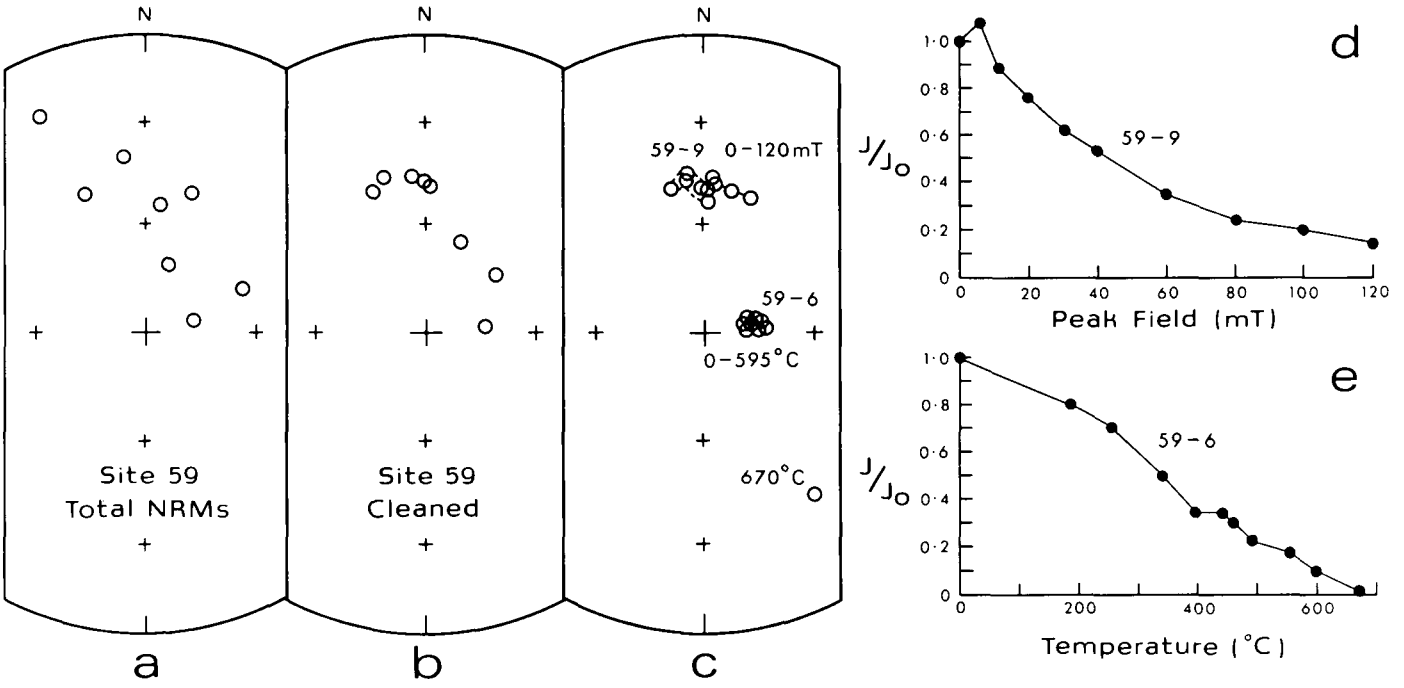



**Figure 7. A site yielding "C" magnetizations: (a) site 59 total NRM directions, (b) site 59 after cleaning at 30 mT (7 samples) and 595 °C (1 sample), (c) to (e) detailed a.f. and thermal demagnetizations of samples from site 59. All projections are equal area: solid circles = lower hemisphere, open circles = upper hemisphere.**

tion) were made by temperatures between 580 and 650 to 690 °C. At the majority of sites, the best grouping of direction was obtained at the NRM stage, and because demagnetization did not generally produce any systematic change in mean direction, the total NRM directions were used to compute the site mean.

### The "C" Direction

The "C" direction of remanence, with northerly declination and moderate to steep negative inclination, was found at 12 sites in various parts of the Belt. At all sites except one, the total NRM directions were already either moderately or well grouped either at or in the vicinity of the "C" direction, and at most of these sites, a.f. demagnetization produced some improvement in the grouping. The demagnetization characteristics of a site which showed good stability to both a.f. and thermal demagnetization are shown in Figure 7. The grains carrying the "C" remanence generally had a wide spectrum of blocking temperatures, from <100 °C to as high as 400 to 600 °C, occasionally with minor contributions to ~650 °C.

### The Other Directions

Eleven sites yielded significant site mean directions which did not fall in any of the three groups of directions just discussed and which appeared to be essentially random in their distribution. At most sites, the total NRM directions were either poorly or moderately well grouped, and demagnetization generally produced a moderate improvement in grouping and only a small change in mean direction. The magnetization showed excellent stability to thermal demagnetization by temperatures up to between 580 and 680 °C.

### Multicomponent Magnetizations

At site 38, both "A" and "B" components were present in every sample and could be separated only be detailed a.f. demagnetization and vector analysis. The site mean total NRM had a fairly typical "B" direction, but progressive a.f. demagnetization caused the site mean observed magnetization to move systematically toward the typical "A" direction and to reach an apparent end point at 40 and 50 mT ~25° from the over-all mean "A" direction (Fig. 8a). Further demagnetizations caused the observed magnetization to move back into the upper hemisphere and to reach an end point at 180 and 200 mT, close to the NRM direction. The site mean vector removed in the NRM/5-mT step is of doubtful significance, as it probably includes recently acquired components. At successively higher steps, however, the removed vectors moved gradually from the typical "B" direction toward the "A" direction, indicating that at first, the "B" component alone was being removed, then an increasing proportion of "A" was removed (Fig. 8b). The 60/80, 80/100, and 100/120-mT removed vectors form an end point reasonably close to the typical "A" direction, suggesting that over these steps, the "A" component alone was being removed. Of course it is impossible to exclude completely the unlikely possibility that over these three steps, the "A" and "B" components were removed in almost exactly constant proportions. The removed vectors above 120 mT were not well grouped within each step, and the site means showed an erratic movement, which, for clarity, is not illustrated, but they were generally scattered about the typical "A" direction. The gradual movement of the observed magnetization back to the "B" direction above 50 mT indicates the presence of a residual high-coercivity "B" component which is not being removed by demagnetization.

These results indicate that at site 38, the "B" component resides in grains with both high and low coercivities, and the "A" component resides in medium coercivity grains (Fig. 8c). Although there is an overlap of the coercivities of the two components between 20 and 60 mT, there is an interval between 60 and 120 mT during which only the "A" component is removed, and it is this that allows the "A" component to be isolated. This situation is in contrast to that revealed by the thermal demagnetization of one

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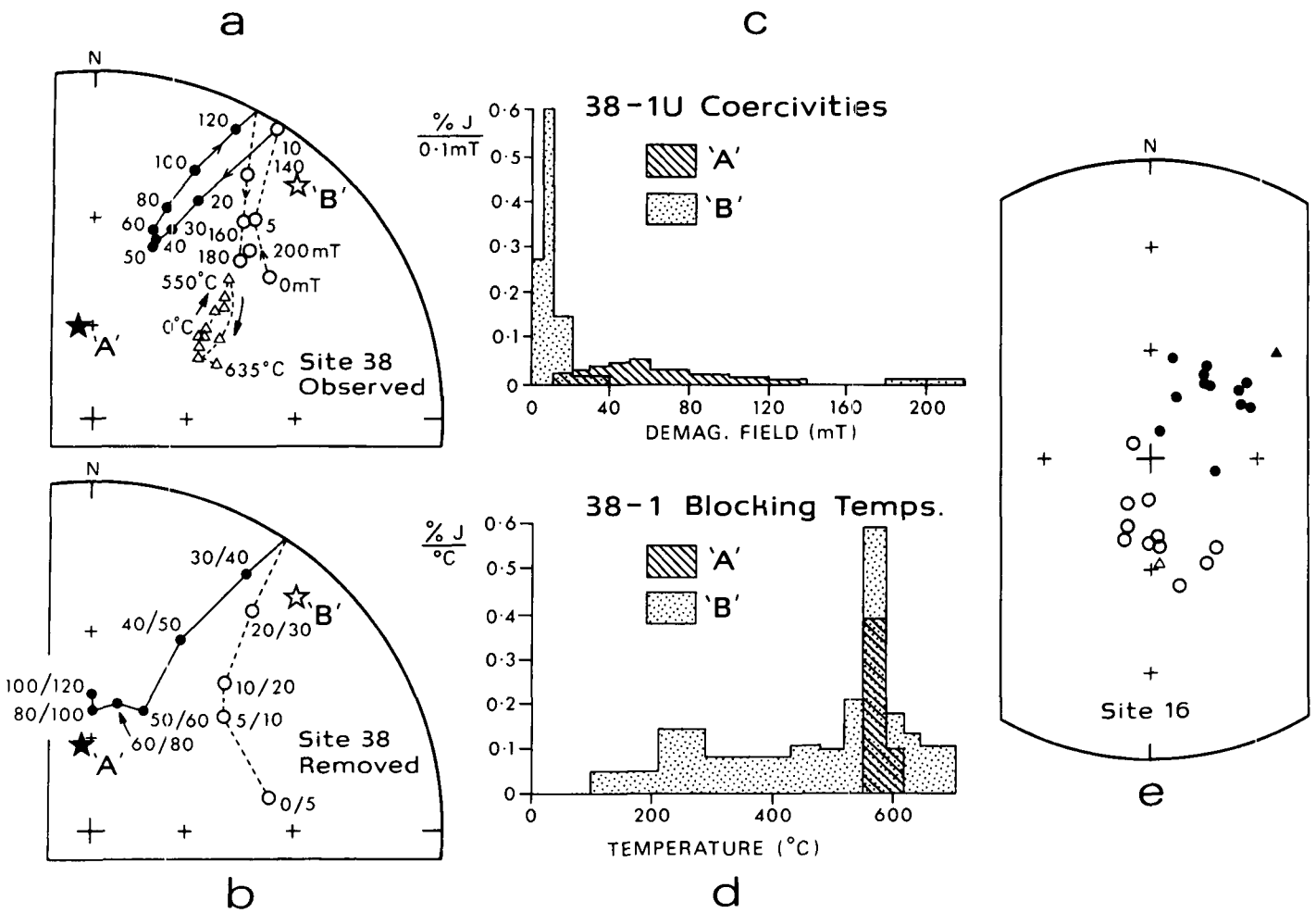



**Figure 8. Multicomponent magnetizations: (a) site 38, movement of site mean observed magnetization with progressive a.f. demagnetization as much as 200 mT (circles), and the movement of one sample during progressive thermal demagnetization (triangles); (b) site 38, site mean removed vectors between 0 and 120 mT. Both (a) and (b) are stereographic projections: solid symbols = lower hemisphere, open symbols = upper hemisphere. Over-all mean "A" and "B" directions from this study are indicated by stars. (c) and (d) site 38, coercivity and blocking temperatures spectra of "A" and "B" components in two samples from the same core; (e) site 16, equal-area projection showing "A" magnetizations in the lower hemisphere revealed by observed magnetization at 100 mT (solid circles) and one removed vector at 545/570 °C (solid triangle). Reversed "A" magnetizations in the upper hemisphere revealed by removed vectors at 20–60 mT (open circles) and one removed vector at 570/623 °C (open triangle).**

sample from site 38 (Fig. 8a). Demagnetization up to 550° caused the observed magnetization to move only ~15° toward the "A" direction; between 550 and 615°, the magnetization moved back to the NRM direction. For all the demagnetization steps except two, the removed vectors (not illustrated) were close to the NRM direction, that is, the "B" component alone was being removed. For the other two steps (550/585° and 585/615°), however, the removed vectors were displaced toward the "A" direction but were still in the upper hemisphere more than 80° from the "A" direction, indicating that the "B" and "A" components were being removed together (in the ratio of ~3:2). The "A" and "B" components at site 38 thus have completely overlapping blocking temperatures spectra (Fig. 8d), and so thermal demagnetization alone cannot isolate the "A" component.

At site 16, both normal and reversed "A" components were found in every sample (Fig. 8e). A high-coercivity component with a fairly typical "A" direction was shown by the observed magnetization after 100-mT demagnetization; after this step, the vectors had reached stable end points and had also achieved their best grouping. The vectors removed during the earlier stages of demagnetization, generally at one or more of the steps between 20 and 60 mT, were also well grouped, and although this low-coercivity component was only approximately antiparallel to the high-coercivity component, it was almost exactly antiparallel to the over-all mean "A" direction. Thermal demagnetization of a single sample from this site showed that the main components were removed over the two steps between 545, 570, and 623 °C. Unexpectedly, it was found that the 545/570 °C removed vector was very close to the high-coercivity direction, and the 570/623 °C removed vector was almost exactly coincident with the low-coercivity direction. The low-coercivity component thus appears to have higher blocking temperatures and the high-coercivity component appears to have lower blocking temperatures!

At site 29 (not illustrated), both "A" and "C" components are present in every sample. The "C" component resides in grains with coercivities corresponding to demagnetizing fields between 0 and 120 mT and in grains with

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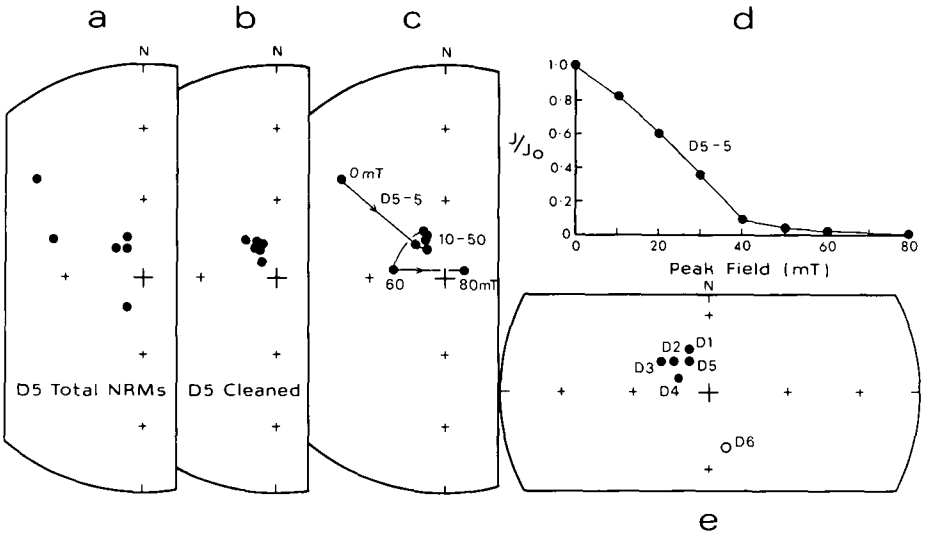



**Figure 9. Post-tectonic Sand River Dykes. (a) site D5 total NRM directions, (b) site D5 after 20 mT cleaning, (c) and (d) detailed a.f. demagnetization of a sample from site D5, (e) cleaned site mean directions for all six dikes. All projections are equal area: solid circles = lower hemisphere, open circles = upper hemisphere.**

coercivities corresponding to demagnetizing fields >180 mT. The "A" component resides in grains with coercivities corresponding to demagnetizing fields between 10 and at least 180 mT. The "A" and "C" components have completely overlapping blocking temperature spectra, and so, as at site 38, the "A" component is isolated by a.f. demagnetization because of a window (here between 120 and 180 mT) where only the "A" component is removed.

## THE POST-TECTONIC SAND RIVER DIKES

Paleomagnetic samples were obtained from 6 undeformed and unmetamorphosed tholeiitic dikes (labeled D1–D6 in Fig. 3) ~9 km southeast of Messina. These appear at the well-exposed pavement in the Sand River ~800 m south of the Messina-Tshipise road. The dikes are all subparallel, striking between 054° and 074°, and are quite closely spaced, the distance between the two outermost dikes being ~150 m. The majority of the dikes are narrow (between 12 and 150 cm wide), but dike D1 is ~20 m wide. Material from dikes D2 and D3 has yielded a Rb-Sr whole-rock isochron of 1876 ± 68 m.y. (Barton, 1979).

Samples from these dikes responded extremely well to a.f. demagnetization and generally reached well-grouped, stable end points at fields of 20–30 mT. One dike, however, (D3), which had extremely scattered total NRM directions with exceptionally high intensities, required fields of as much as 120 mT before stable end points were obtained. The total NRM distribution and a.f. demagnetization behavior of a typical dike is shown in Figures 9a–9d. Five of the dikes yielded cleaned site mean directions which were very closely grouped with northwest declination and steep positive inclination, and the remaining dike gave a site mean which was almost exactly reversed with respect to this group (Fig. 9e). Site statistics are given in Appendix 1, and over-all site mean statistics are given in Table 1.

TABLE 2. PALEOMAGNETIC RESULTS FROM THE KALAHARI CRATON FOR THE INTERVAL CIRCA 2050–1800 m.y.

| Rock unit | Abbreviation (Fig. 10) | Age (m.y.) | No. of sites | Paleomagnetic pole | Reference |
|---|---|---|---|---|---|
| MAJOR STUDIES | | | | | |
| Bushveld Gabbro | BG | 2096 ± 12[f]<br>2051 ±24[a] | 5 | 23°N, 36°E ($A_{95}$ = 12°) | 2, 3, 4 |
| Palabora I | PAL 1 | 2047 ± 11/–8[g] | 11 | 35°N, 45°E (dp = 7°, dm = 11°) | 5, 6, 7 |
| Palabora II | PAL 2 | 2012 ± 19[h] | 10 | 4°N, 356°E (dp = 9°, dm = 11°) | |
| Vredefort Complex | VC | 2006 ± 47[b]<br>1938b<br>1950 ± 60[a] | 8 | 26°N, 39°E ($A_{95}$ = 12°) | 8, 9, 10 |
| Limpopo Metamorphics "A" | | 1980 ± 39[e] | 21 | 26°N, 22°E (dp = 8°, dm = 10°) | 1, 11, 12 |
| Great Dyke Extensions | GDE | 1895 ± 73[d] | 7 | 20°N, O°E (dp = 12°, dm = 15°) | 13, 11 |
| Sand River Dykes | | 1876 ± 68[a] | 6 | 3°N, 9°E (dp =10°, dm =11°) | 1, 14 |
| Mashonaland Dolerites | MD | 1811 ± 20[c] | 14 | 6°N, 339°E (dp =5°, dm = 6°) | 15, 16 |
| MINOR STUDIES (<5 SITES) | | | | | |
| Losberg Gabbro | LG | 1841 ± 276 | 1 | 33°N, 36°E | 8, 17 |
| Waterberg Redbeds | | | | | |
| sites J, K, M | W1 | bracketed between 2000–1350 (certainly) and 2000–1720 (possibly) | 3 | 36°N, 48°E (dp = 15°, dm = 24°) | 18, 19 |
| site D | W2 | " | 1 | 8°N, 10°E | " |
| sites F, G, L, I | W3 | " | 4 | 3°N, 333°E (dp =14°, dm = 21°) | " |
| Rust der Winter Volcanic | RDW | 1790 ± 70 | 1 | 16°N, 35°E | 5, 20 |

*Note:* radiometric dating details: a = Rb-Sr WR iso.; b = Rb-Sr min. iso.; c = Rb-Sr WR-min. iso.; d = mean of 5 Rb-Sr min. cooling ages from nearby gneisses; e = mean of 22 Rb-Sr min., min.-pair, min.-WR, and thin-slice ages; f = $^{40}Ar/^{39}Ar$ WR; g = U-Pb concordia using uranothorianites and baddeleyites; H = Rb-Sr phlogopite iso.

References cited: 1. This study; 2. Gough and Van Niekerk (1959); 3. Hamilton (1977); 4. Coertze and others (1978); 5. Morgan and Briden (1981); 6. Eriksson (1982); 7. Eriksson (in press); 8. Hargraves (1970); 9. Nicolaysen and others (1963); 10. Hart and others (1981); 11. Van Breemen and Dodson (1972); 12. Hickman and Wakefield (1975); 13. Jones and others (1975); 14. Barton (1979); 15. McElhinny and Opdyke (1964); 16. Compston and McElhinny (1975); 17. Davies and others (1970); 18. Jones and McElhinny (1967); 19. McElhinny (1968); 20. Oosthuyzen (1964).

## DISCUSSION

### The Origin and Age of the Magnetizations

It has been known for some time that the apparent polar wander (a.p.w.) path for the Kalahari Shield (essentially the Kaapvaal and Rhodesian cratons, together with the Limpopo Belt) for the interval circa 2050 to 1800 m.y. passes from east to west across northern Africa (Fig. 10; Table 2). The paleomagnetic pole corresponding to the predominant "A" magnetization in the Limpopo Belt metamorphic rocks falls squarely on this path at a point corresponding to an age of ~2000–1950 m.y. It is inferred, then, that this magnetization is most probably thermoremanent magnetization (TRM) acquired during the cooling episode reflected by the widespread (circa 1980 m.y.) mineral ages discussed above. The pole from the post-tectonic Sand River Dykes falls very slightly outside the path at a point corresponding to an age of ~1850–1900 m.y. This correlates well with their Rb-Sr whole-rock isochron age of 1876 m.y. and suggests that their magnetization was acquired at the time of intrusion. Because the Sand River Dykes pole was obtained from closely spaced parallel dikes at the same locality, there must be some doubt as to whether the

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Dykes were intruded over a sufficient time span for any effects of secular variation to be averaged out. The fact, however, that one of the Dykes is reversed with respect to the remainder indicates that they were not all emplaced at the same instant. These new paleomagnetic data reported here, together with the previously published data from the Kalahari Shield, result in a particularly well defined a.p.w. path segment ~80° long and about ±10° lateral uncertainty for the interval circa 2050–1800 m.y.

If the identification of the "A" magnetization as a TRM acquired during the ~1980-m.y. cooling episode is correct, then the characteristic blocking temperature spectra of the "A" and "B" magnetizations (exemplified by their spectra when they occur together at site 38; see Fig. 8) place constraints on the possible modes of origin of the "B" magnetization. "B" must be younger than "A," otherwise the lower blocking temperature components of "B" would have been reset by the ~1980-m.y. event. Moreover, "B" cannot be the result of a later, simple (that is, TRM) thermal overprint, as this would not explain the "B" components with blocking temperatures higher than those of "A." The most plausible explanation is that the "B" magnetization is a chemical remanent magnetization (CRM) acquired some time after 1980 m.y. B.P., chemical growth having produced grains with a very wide range of blocking temperatures. In the absence of any appreciable later event with which to correlate it, the age of the "B" magnetization remains uncertain at present. It is simply noted in passing that the pole corresponding to the "B" magnetization (reversed) falls on the a.p.w. path swath for Africa (McWilliams and Kröner, 1981) at a point corresponding to an age of 750–800 m.y.

The wide range of blocking temperatures shown by the "C" magnetizations places constraints on the age of this magnetization that are similar to those applied to the "B" magnetization. The mean "C" direction is statistically indistinguishable (at the 95% probability level) from the present Earth's field direction in the Limpopo Belt (D = 345°, I = –60°), and so it is probable, but it cannot be proven, that the "C" magnetization is of very recent origin. Note that this present field deviates considerably from the axial dipole field. This is reflected in the fact that the oval of 95% confidence around the paleomagnetic pole from the "C" magnetization does not cover the present geographic pole. The presence, at some sites, of very high blocking temperature components in the "C" magnetizations suggests that the magnetization is at least partly of chemical origin, that is, it reflects incipient weathering.

### Diachronous Cooling of the Limpopo Belt

It was noted above (see section entitled Geochronology) that the evidence of the Rb-Sr mineral ages indicates that whereas the main part of the Belt (corresponding quite closely to the sampling area of the present study) cooled at ~1980 m.y. B.P., the northern part of the Belt, in the region of the extensions of the Great Dyke and its satellites into the Belt, cooled ~80 m.y. later, at ~1895 m.y. B.P. Although it was not possible to sample the metamorphic rocks in this northern part of the Belt during this study, paleomagnetic results have been reported (Jones and others, 1975) from seven sites in the Great Dyke extensions into this part of the Belt (sampling sites shown in Fig. 3 by open circles). These dikes have yielded a Rb-Sr whole-rock isochron of 2550 m.y. (Robertson and Van Breemen, 1970), similar to the age of the Great Dyke in the Rhodesian craton. Presumably, these dikes experienced the cooling event at ~1895 m.y. B.P., as recorded by the mineral ages in the gneisses in this part of the Belt. The mean pole for the Great Dyke extensions (Table 2; Fig. 10) is significantly different from the Great Dyke pole. It falls on the a.p.w. path at a point corresponding to an age of ~1900–1950 m.y., suggesting that the magnetization in these dikes was, in fact, set (or reset) in this regional cooling event.

The Great Dyke extensions pole is displaced ~20° along the a.p.w. path in a direction that is younger than the Limpopo Metamorphics "A" pole reported here. The paleomagnetic results thus reinforce the radiometric evidence that the more northerly part of the Belt cooled appreciably later than did the more southerly and westerly parts. Furthermore, the Great Dyke extensions and Limpopo Metamorphics "A" poles are of opposite polarity. All of the site poles making up the mean Great Dyke extensions pole are magnetic south poles, whereas 19 out of the 21 site poles making up the over-all Limpopo Metamorphics "A" pole are magnetic north poles. [As pointed out above, we cannot be certain that the two south poles in the "A" group (sites 16a and 53) are not, in fact, part of the population of sites with random directions and so perhaps should not properly be considered as "A" magnetizations.] During the time interval between the cooling of the two parts of the Belt, there was thus a change in geomagnetic field polarity. Because these rocks almost certainly acquired their magnetizations very slowly, and hence contain a record of the magnetic field averaged over millions or possibly tens of millions of years (Morgan, 1976), it is not possible at present to say whether this change was a reversal from one constant polarity to the other or merely a change in polarity bias. There is the fascinating possibility that more detailed sampling of the metamorphic rocks of the Limpopo Belt, especially in the so far unsampled parts of the Belt in Zimbabwe, might allow this polarity change to be mapped on the ground.

The difference in cooling ages between the two parts of the Belt could be explained in several different ways. The simplest explanation would be that whatever produced the cooling, whether it was the decay of a regional thermal pulse or uplift and erosion, simply occurred earlier in the southerly and westerly parts of the Belt than in the northerly part. A possible alter-

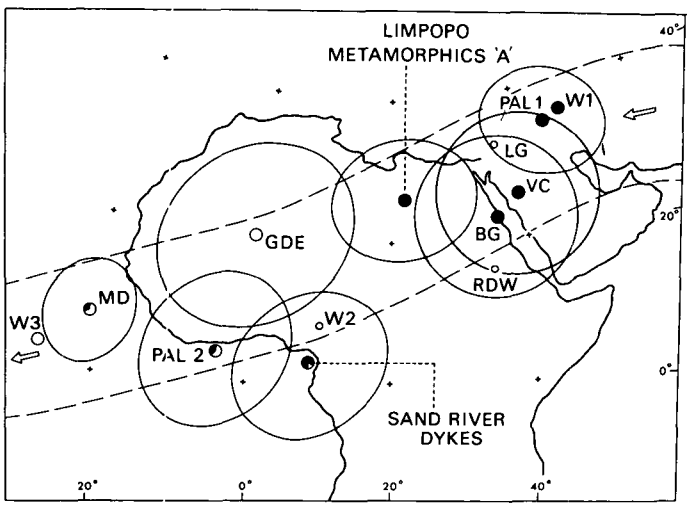



**Figure 10. Apparent polar wander path for the Kalahari Shield for the interval circa 2050–1800 m.y. B.P. The abbreviations for previously published results are explained in Table 2. Poles from major studies are shown with an oval of confidence, poles from minor studies without an oval, and poles based on only one site are represented by small circles. Solid circles represent N poles; open circles, S poles; and partly filled circles, mixed polarities in the proportion indicated.**

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nate explanation follows from the fact that, however it may have been caused, the cooling will have almost certainly involved the very slow downward movement, relative to the crustal block, of approximately horizontal isotherms. This will have resulted in the higher crustal levels acquiring the oldest magnetizations and mineral ages and successively deeper levels acquiring younger magnetizations and mineral ages (Morgan, 1976). The difference in magnetization and mineral ages could thus have arisen, because some time after final cooling of the Belt, there were relative vertical movements which have resulted in the present erosion surface in the northern area representing a crustal level that was originally deeper than the present erosion surface in the southern and western areas. The relative vertical movements could have been faulting (the northern area upfaulted relative to the rest of the Belt) or slight tilting (with a dip toward the south and east). If this explanation is correct, the original cooling could have been synchronous across the Belt. It is, of course, also possible that both effects, that is, the original diachronous cooling across the Belt and also later relative vertical movements, could have contributed to the presently observed cooling pattern seen in the Belt.

It is noted that in their study of the deep structure of the Limpopo Belt, using surface structural geology and gravity data, Coward and Fairhead (1980) also demonstrated the existence of a north-south asymmetry across the Belt. It is likely, however, that most of the movement, especially vertical movement, which resulted in the asymmetry described by Coward and Fairhead, occurred considerably earlier than did the cooling episode being discussed here. It is therefore unlikely that these movements were related to the observed cooling pattern. Determining whether the northern and southern/western parts of the Belt represent different crustal levels is made difficult by the fact that we are concerned with the crustal level (that is, depth) at the time of final cooling and not at the peak of metamorphism. Because of the considerable amount of relative movement that occurred within the Belt prior to final cooling (Coward and Fairhead, 1980), metamorphic grade in the Belt is therefore no clue to crustal level at the time of final cooling. The amount of tilting that would be necessary to produce the required difference in crustal level across the width of the belt is so small (probably $<2°$) that it is undetectable in this type of terrain. It is noted, however, that within the Belt, there are several major east-west fault systems, for example, the Tuli-Sabi fault system and the Soutpansberg fault zone. These systems show evidence of considerable post-2000-m.y. vertical movements with the correct sense, that is, downthrown to the south (Mason, 1973).

TABLE 3. PALEOMAGNETIC RESULTS FROM AFRICA OUTSIDE THE KALAHARI SHIELD AND FROM SOUTH AMERICA FOR THE GENERAL INTERVAL 2050–1800 m.y.

| Rock Unit | Abbreviation (Fig. 11) | Age (m.y.) | No. of sites | Pole Lat. (°N) | Pole Long. (°E) | $A_{95}$ | Reference |
|---|---|---|---|---|---|---|---|
| WEST AFRICAN SHIELD* | | | | | | | |
| Aftout Gabbro | AG | Bracketed between 1940 ± 50[a] 1908 ± 50[a] | 5 | 29 | 55 | 6 | 1,2,3 |
| Basic Yetti Dykes | BYD | " | 7 | 41 | 31 | 33 | " |
| Aftout Plutons "C" | APC | " | 4 | ~5 | 93 | 11 | " |
| Aftout Plutons "E" | APE | " | 6 | ~6 | 90 | 5 | " |
| RICHTERSVELD PROVINCE | | | | | | | |
| Orange River Lavas | ORL | 2067 ± 83[f] 1958 ± 61[g] 1957 ± 54[b] | 3 | −19 | 74 | 20 | 4,5 |
| GUAYANA SHIELD | | | | | | | |
| Supamo-Pastora | SP | 2000–1900[d] | 2 | −50 | 346 | 98 | 6 |
| Imataca Pole 1 | $I_1$ | 2055–2000[a,c,e] | 9 | −49 | 18 | 9 | " |
| Imataca Pole 2 | $I_2$ | " | 4 | −29 | 21 | 13 | " |

*Note:* radiometric dating details: a = Rb-Sr; b = Rb-Sr WR iso.; c = Rb-Sr biotite; d = K-Ar hornblende and biotite; e = U-Pb; f = Pb-Pb WR iso.; g = Th-Pb WR iso.

References cited: 1. Lomax (1975); 2. Sabaté and Lomax (1975); 3. McElhinny and Cowley (1980); 4. Piper (1975); 5. Reid (1979); 6. Onstott and Hargraves (1981).

*Note that the Aftout Diorite pole quoted by Sabaté and Lomax (1975) and the seven poles from this unit given by Piper (1976) have not been included here, as the unit was later (Lomax, 1975) recognized as being essentially randomly magnetized.

The change in polarity between the Limpopo Metamorphics "A" pole and the Great Dyke extensions pole is simply part of a larger polarity pattern shown by the 2050–1800-m.y. a.p.w.-path segment. The older, eastern part of the path is made up almost exclusively of north poles; the younger, western part of the path is of mixed polarity but with a suggestion of a predominance of south poles. The transition between the two polarity elements occurs between the Limpopo Metamorphics "A" and the Great Dyke extensions poles. It may be significant that of the four major poles in the mixed polarity part of the path, only the Great Dyke extensions pole is made up exclusively of south poles, whereas the poles for the Sand River Dykes, Mashonaland Dolerites, and Palabora II (also consisting mainly of dikes) are of mixed polarity. It is possible that the Great Dyke extensions magnetizations, almost certainly acquired during slow regional cooling, are a time-averaged record of a geomagnetic field, the detailed behavior of which is recorded in the spot-reading magnetizations of the other three studies.

## Comparison of Kalahari Paleomagnetic Results with Those from Other African Shields, South America, and North America

Paleomagnetic results from units with at least moderately good age control from Africa outside the Kalahari Shield (West African Shield and Richtersveld Province) and from South America (Guayana Shield) for the interval circa 2050–1800 m.y. are listed in Table 3 and illustrated in Figure 11, which also shows the a.p.w. path for the Kalahari Shield for comparison. Of the five African results listed, two are based on studies involving less than five sites, and two of the poles have an associated $A_{95}$ of 20° or more. It should also be noted that all four results from the West African Shield were obtained from the same general locality in western Algeria. The Aftout Gabbro pole falls on the a.p.w. path for the Kalahari Shield, and the Basic Yetti Dykes pole falls very close to the Kalahari path, but the two remaining West African poles, Aftout Plutons "C" and "E," fall well to the southeast of the path. The pole from the Orange River Lavas (Richtersveld Province) also falls well to the southeast of the path in the same general region as the two Aftout Plutons poles. Of the three results from South America, two, Imataca 1 and 2, appear to be reasonably well defined, but the Sumapo-Pastora result is based on only two sites and has an extremely large $A_{95}$. From Figure 11, it can be seen that on the Bullard and others (1965) reconstruction, these three South American results plot at fairly scattered localities well to the southeast of the Kalahari path in approximately the same position as some of the African results.

Onstott and Hargraves (1981), using a larger and somewhat different data base, suggested that a better fit of the African and South American data is obtained if the Guayana Shield (and an area to the south and west) is displaced ~1500 km southward relative to the rest of South America. From the more limited data base considered here, there would appear to be no evidence for this. Onstott and Hargraves also suggested that a better fit between the Kalahari results and the other African and the South American results is obtained if the Kalahari Shield and neighboring parts of southern Africa are rotated, relative to the rest of Africa, 64° clockwise about a pole at 44°E, 28°S. This rotation of the Kalahari region and its a.p.w. path is shown by dashed lines in Figure 11, and although it produces some improvement (five non-Kalahari poles fall on or very close to the path, as opposed to two before rotation), the

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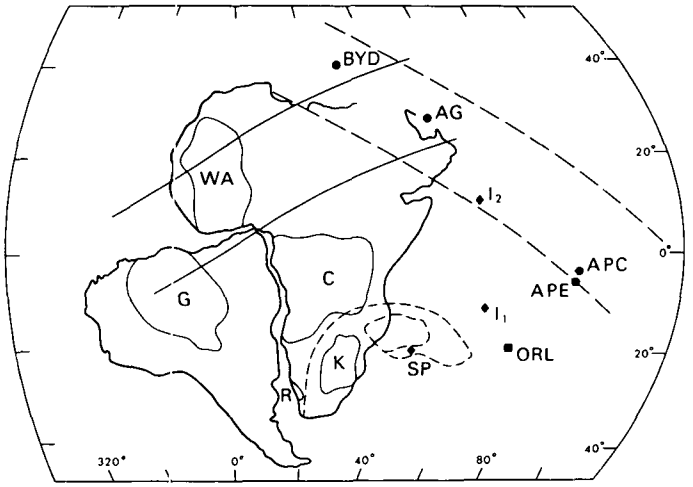




**Figure 11. Paleomagnetic results from Africa outside the Kalahari Shield and from South America for the interval circa 2050–1800 m.y. B.P. South America and its poles have been shifted relative to Africa according to the reconstruction of Bullard and others (1965). Poles from the West African Shield (WA) - circles, Richtersveld Province (R) - squares, and Guayana Shield (G) - diamonds. Pole abbreviations are explained in Table 3. The a.p.w. path for the Kalahari Shield (K) is also shown (solid lines) for comparison. The Kalahari Shield and its a.p.w. path are also shown (dashed lines) after the rotation relative to the rest of Africa as suggested by Onstott and Hargraves (1981).**

improvement is not sufficient to provide strong evidence in favor of this reconstruction. In fact, on the basis of the present evidence, it would appear to be difficult to draw any firm conclusions about the relative positions of any of these shield areas at this time.

There has been considerable discussion recently concerning the arrangement of the continents in the Proterozoic; in particular, Piper (1976, 1978, 1980) interpreted the paleomagnetic evidence as indicating the existence of a large supercontinent for much of Proterozoic time. Irving and McGlynn (1979) opposed this view, citing as part of their evidence the contrasting shape and position on the proposed reconstruction of the African and Laurentian (North American) a.p.w. paths during the period 2000–1750 m.y. B.P. The improved definition of the Kalahari a.p.w. path for the interval circa 2050–1800 m.y. B.P. now obtained allows a more reliable test of the relative positions of Africa and Laurentia at this time (Fig. 12). Part of the disagreement has centered on the fact that there have been different interpretations of the Laurentian a.p.w. path during this period, the two main versions being the ones given by Irving and McGlynn (1979) (Fig. 12a) and by Piper (1980) (Fig. 12b). If one puts Africa and Laurentia in their supercontinent positions, the Irving and McGlynn Laurentian path and the Kalahari path are approximately coincident at ~2050 m.y. B.P. but thereafter are widely divergent in position and considerably different in shape. Surprisingly, the Laurentian path of Piper, although coincident with the Kalahari path at ~2050 and 1800 m.y. B.P., is totally different from the Kalahari path in both position and shape in the intervening period. Neither interpretation of the Laurentian paleomagnetic data over this time interval thus gives any support to the supercontinent reconstruction of Africa and Laurentia.

More recently, Piper (1982) reaffirmed his support for a Proterozoic supercontinent with Africa and Laurentia in the same relative positions as in the previous reconstructions. In particular, he shows (Piper, 1982, Fig. 3) a single a.p.w. path passing through the 2000–1750-m.y. B.P. paleomagnetic poles from essentially all Precambrian Shield areas, including Africa and Laurentia. The 1900–1800-m.y. B.P. part of this path accords well with the African data shown and corresponds fairly closely with the 2050–1800-m.y. B.P. Kalahari path presented in the present paper. The Laurentian data, however, do not show a good fit with this part of the path, especially the sequence of poles numbered 4 to 10 which appear to define a southerly loop similar to that present in Piper's 1980 path but not present in this 1982 version.

On the basis of this evidence concerning the Kalahari Shield and Laurentia, and considering the inconclusive evidence presented above regarding the relative positions of the other African shields, it is concluded that over the time interval circa 2050–1800 m.y. B.P., there is no paleomagnetic support for the proposed supercontinent reconstruction of Africa and Laurentia. In fact, the evidence strongly suggests that at least the Kalahari Shield moved substantially with respect to Laurentia at this time.

## ACKNOWLEDGMENTS

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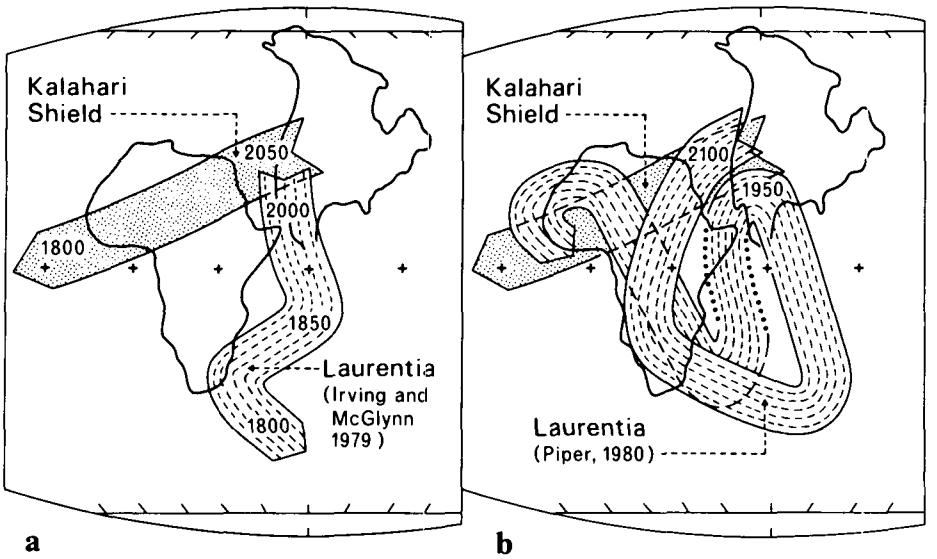



**Figure 12. The a.p.w. path for the Kalahari Shield for the interval circa 2050–1800 m.y. B.P. This is compared with the a.p.w. path for Laurentia (North America) for the same time period after Laurentia and its a.p.w. path have been shifted relative to Africa, according to the Proterozoic supercontinent reconstruction of Piper (1976 and 1982). The Laurentian a.p.w. path is shown in (a) according to Irving and McGlynn (1979) and in (b) according to Piper (1980).**

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sampling in the Messina region, and Dr. R. M. Key, of the Geological Survey of Botswana, gave advice on sampling localities in Botswana. Dr. C. A. Foss gave field assistance during the first field season. The final paper benefited greatly from critical reviews by Dr. T. C. Onstott and Professor R. B. Hargraves. The diagrams were drawn by Mr. R. Boud. The study was funded by the University of Leeds and the United Kingdom Natural Environment Research Council.

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APPENDIX 1. PALEOMAGNETIC RESULTS FROM LIMPOPO BELT SITES YIELDING USEFUL MAGNETIZATIONS

| Site | Lithology | Cleaning | N | n | D | I | k | $\alpha_{95}$ |
|---|---|---|---|---|---|---|---|---|
| 1. METAMORPHIC ROCKS | | | | | | | | |
| Group "A" magnetizations | | | | | | | | |
| 3 | feld-q-horn-biot gneiss | 80–140 mT/560°C | 7 | 3 | 351.0 | 49.2 | 31.8 | 10.8 |
| 10 | horn-feld-q-biot gneiss | 15 mT/450°C | 6 | 1 | 348.7 | 59.2 | 12.9 | 19.3 |
| 15 | feld-q-cpx-biot gneiss | 140–160 mT/635°C | 11 | ·· | 6.2 | 58.2 | 147.0 | 3.7 |
| 16(a) | feld-q-opx-(cpx) gneiss | removed vectors 20–60 mT/ removed vectors 570–623°C | 12 | 1 | 175.4 | –62.9 | 23.8 | 9.0 |
| 16(b) | feld-q-opx-(cpx) gneiss | observed magnts. 100 mT/ removed vectors 545–570°C | 13 | ·· | 43.6 | 57.0 | 28.7 | 7.8 |
| 22 | feld-q-(biot) gneiss | NRM | 6 | 2 | 334.5 | 77.1 | 31.8 | 12.0 |
| 29 | feld-q-(horn) gneiss | removed vectors 120–180 mT | 9 | ·· | 353.5 | 66.1 | 133.9 | 4.4 |
| 31 | feld-q-(biot) gneiss | 40–100 mT | 9 | 6 | 6.3 | 68.0 | 23.0 | 10.9 |
| 32 | feld-q gneiss | 90 mT/585°C | 8 | 1 | 342.0 | 51.5 | 88.6 | 5.9 |
| 33 | feld-q-horn-opx gneiss | 120–180 mT | 9 | ·· | 323.2 | 55.7 | 51.7 | 7.2 |
| 34 | feld-q-(horn) gneiss | 60 mT | 5 | 2 | 10.3 | 56.5 | 21.2 | 16.9 |
| 38 | feld-q-horn gneiss | removed vectors 60–120 mT | 8 | ·· | 5.2 | 48.2 | 42.8 | 8.5 |
| 48 | meta-andesite (Tati Schist Relict) | 20 mT/570°C | 9 | ·· | 1.0 | 58.2 | 138.2 | 4.3 |
| 53 | feld-q-biot gneiss | 10–80 mT | 9 | ·· | 159.0 | –70.0 | 14.2 | 14.1 |
| 57 | feld-q-opx-biot gneiss | 20 mT/569°C | 9 | 1 | 0.4 | 50.5 | 40.7 | 8.1 |
| 58 | diorite (Tati Schist Relict) | 140–180 mT | 6 | 3 | 352.8 | 62.5 | 60.9 | 8.6 |
| 60 | amphibolite (Tati Schist Relict) | 520°C | 7 | ·· | 273.8 | 67.9 | 62.1 | 7.7 |
| 65 | feld-q-biot gneiss | 80–120 mT | 7 | 1 | 339.2 | 33.3 | 99.4 | 6.0 |
| 69 | garnet-q-feld-biot gneiss | 30–140 mT | 8 | ·· | 333.1 | 77.4 | 34.7 | 9.5 |
| 70 | feld-q-biot-horn gneiss | 540–557°C | 8 | ·· | 341.6 | 26.9 | 26.6 | 10.9 |
| 86 | granite | 10–60 mT | 8 | ·· | 16.2 | 64.7 | 109.5 | 5.3 |
| Group "B" magnetizations | | | | | | | | |
| 11 | calc-dol-serp (calc silicate) | NRM | 7 | ·· | 53.9 | –7.3 | 27.2 | 11.7 |
| 25 | feld-q-biot gneiss | 20 mT/562°C | 10 | ·· | 55.5 | –4.7 | 15.8 | 12.5 |
| 35 | feld-q-gneiss | 40 mT | 5 | 2 | 32.9 | –4.4 | 18.2 | 18.3 |
| 37 | feld-q-(biot) gneiss | NRM | 6 | 3 | 37.8 | –27.3 | 346.5 | 3.6 |
| 38 | feld-q-horn gneiss | removed vectors 5–20 mT and 180–200 mT | 8 | ·· | 43.7 | –29.1 | 65.3 | 6.9 |
| 42 | feld-q-biot gneiss | NRM | 10 | ·· | 26.6 | –23.7 | 31.3 | 8.7 |
| 45 | feld-q-biot gneiss | NRM | 7 | 3 | 33.7 | 26.1 | 62.7 | 7.6 |
| 52 | feld-q-biot-(sphene) gneiss | NRM | 6 | 3 | 43.3 | 20.1 | 140.2 | 5.6 |
| Group "C" magnetizations | | | | | | | | |
| 26 | feld-q-horn gneiss | 40 mT/577°C | 9 | ·· | 345.2 | –31.3 | 36.7 | 8.6 |
| 28 | feld-q-(biot) gneiss | 30 mT/459°C | 9 | 1 | 347.4 | –51.0 | 52.4 | 7.1 |
| 29 | feld-q-(horn) gneiss | removed vectors 5–10 mT/ observed magnt. 635°C | 9 | ·· | 30.3 | –53.7 | 77.1 | 5.8 |
| 49 | horn-feld-q gneiss (Tati Schist Relict) | 20 mT | 9 | ·· | 351.0 | –67.4 | 8.3 | 18.9 |
| 59 | amphibolite (Tati Schist Relict) | 30 mT/595°C | 8 | ·· | 4.1 | –57.4 | 17.7 | 13.5 |
| 61 | horn-feld-q gneiss (Tati Schist Relict) | NRM | 10 | ·· | 345.8 | –60.7 | 31.8 | 8.6 |
| 62 | feld-q-biot gneiss | 10 mT/394°C | 12 | ·· | 13.1 | –68.3 | 9.3 | 15.0 |
| 64 | feld-q-biot gneiss | 5 mT | 7 | ·· | 56.5 | –77.3 | 12.6 | 17.6 |
| 68 | feld-q-biot gneiss | 40 mT | 7 | 2 | 308.0 | –64.2 | 13.3 | 17.1 |
| 73 | cpx-feld-scap-calc (calc silicate) | 10 mT/342°C | 9 | ·· | 352.8 | –83.1 | 23.5 | 10.8 |
| 76 | feld-q-horn-biot gneiss | NRM | 9 | ·· | 17.6 | –71.8 | 24.2 | 10.6 |
| 78 | biot-feld-q gneiss | 5 mT | 6 | 2 | 334.8 | –87.1 | 12.9 | 19.3 |
| Other magnetizations | | | | | | | | |
| 6 | feld-q-horn-biot gneiss | 120 mT | 7 | 1 | 315.8 | –16.1 | 20.4 | 13.6 |
| 21 | feld-biot-q gneiss | 20–40 mT | 8 | ·· | 279.1 | 15.2 | 6.3 | 23.7 |
| 19 | feld-q-(horn) gneiss | 100 mT/586°C | 8 | ·· | 134.3 | 2.1 | 12.3 | 16.3 |
| 24 | feld-q-(horn) gneiss | 20 mT | 5 | 4 | 252.0 | –35.1 | 260.2 | 4.7 |
| 30 | feld-q-(biot-garnet-opx) gneiss | NRM | 9 | ·· | 186.6 | 0.9 | 12.6 | 15.0 |
| 39 | feld-q-horn gneiss | 10 mT/570°C | 9 | ·· | 222.5 | 44.4 | 36.4 | 8.6 |
| 40 | feld-q-horn-biot gneiss | NRM | 8 | 2 | 195.2 | –17.8 | 10.2 | 18.1 |
| 41 | feld-q-biot gneiss | 100 mT | 7 | 2 | 72.8 | 10.5 | 12.7 | 17.5 |
| 63 | feld-q-biot gneiss | 5 mT/539°C | 6 | 3 | 107.1 | –56.5 | 81.9 | 7.4 |
| 85 | feld-garnet (anorthosite) | 40 mT | 8 | 3 | 102.9 | –36.4 | 12.9 | 16.0 |
| 87 | garnet-biot-feld-q gneiss | NRM | 7 | 1 | 209.6 | –43.4 | 26.2 | 11.9 |
| 2. POST-TECTONIC SAND RIVER DYKES | | | | | | | | |
| D1 | dolerite | 30 mT | 5 | ·· | 332.1 | 71.4 | 56.0 | 10.3 |
| D2 | " | 30–40 mT | 8 | ·· | 306.6 | 70.9 | 171.1 | 4.2 |
| D3 | " | 30–120 mT | 6 | ·· | 301.7 | 67.8 | 103.5 | 6.6 |
| D4 | " | 20 mT | 8 | ·· | 292.5 | 77.0 | 632.8 | 2.2 |
| D5 | " | 20 mT | 6 | ·· | 323.8 | 75.7 | 904.2 | 2.2 |
| D6 | " | 20 mT | 8 | ·· | 163.5 | –67.3 | 973.4 | 1.7 |

*Note:* Where both a.f. and thermal cleaning are indicated, generally only one sample has been cleaned thermally and the remainder have been cleaned with a.f. N is the number of samples used in the statistics, n is the number of rejected samples (see text), D and I are the declination and inclination of the mean direction, k is Fisher's precision parameter, and $\alpha_{95}$ is the radius of 95% confidence about the mean direction. The lithology abbreviations are: feld = feldspar, q = quartz, biot = biotite, horn = hornblende, cpx = clinopyroxene, opx = orthopyroxene, calc = calcite, dol = dolomite, serp = serpentie, scap = scapolite.

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